IndustryGPT: Parameter-Efficient Fine-Tuning of Large Language Models for Specialized Information Technology Conversational Agents.

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2. Abstract

Large Language Models (LLMs) have ushered in a revolutionary, paradigm-shifting trend in the domain of natural language processing, showcasing unprecedented and highly versatile capabilities in the realms of text generation, summarization, and zero-shot reasoning. However, leveraging these basic and generalized models for highly specialized, syntax-centric domains such as Information Technology (IT) architecture and software engineering is fraught with substantial computational, logistical, and economic challenges. Fine-tuning large models necessitates a high degree of hardware infrastructure, effectively creating a barrier for localized, offline, or resource-constrained development. This paper presents the concept of IndustryGPT, an internally hosted and highly specialized conversational AI designed specifically to meet the stringent needs of the highly competitive and demanding IT sector. Utilizing the highly regarded pacovaldez/stackoverflow-questions dataset, we utilized the most advanced Parameter Efficient Fine Tuning techniques to fine-tune the highly regarded and powerful GPT-2 model. The implementation utilizes the highly effective Low Rank Adaptation technique, which allows the insertion of trainable rank decomposition matrices into the transformer model, while the fundamental weights are frozen. The methodological focus of this research heavily emphasizes the concept of resource-constrained, democratized AI development, as we successfully train the model utilizing a single, consumer-grade NVIDIA T4 GPU, effectively demonstrating the concept of domain adaptation without the need for massive, enterprise-grade computational resources. The resultant model exhibits a highly developed understanding of complex software engineering, database optimization, and architectural paradigms.

Keywords: Large Language Models, Parameter-Efficient Fine-Tuning, Low-Rank Adaptation (LoRA), Information Technology, Conversational Agents, Natural Language Processing, Resource-Constrained Machine Learning.

3. Introduction

- 3.1 **Background and Context**
- The introduction of deep learning models, specifically the self-attention mechanism popularized in the Transformer architecture, has catalyzed unprecedented, revolutionary progress in the creation of automated agents of conversation. The foundational models, including OpenAI's GPT-3, Meta's LLaMA, and a number of its open-source derivatives, have been trained on a massive, generalizable corpus of text, including a substantial percentage of the public internet. Such a broad scope of training enables them to perform with unprecedented excellence in a broad range of general, multi-disciplinary tasks. However, the modern business environment is characterized by an unprecedented need for hyper-specific domains. The domain of Information Technology, including DevOps, database administration, software engineering, and systems architecture, demands an unprecedented degree of exactness, which general language models are unable to provide.

3.2 The Research Problem

- Despite the unprecedented versatility of general language models, they frequently exhibit critical shortcomings in highly technical, domain-specific environments. Specifically, general models often "hallucinate" when presented with a complex debugging or systems architecture question, providing technically incorrect, albeit semantically plausible, responses. Moreover, general models often lack the precise technical vocabulary or understanding of the highly specific context of overlapping programming paradigms and modern systems architecture. To effectively overcome the aforementioned shortcomings of general language models, a model must be strictly grounded in domain-specific data. Traditionally, this has been accomplished through a process of fine-tuning a model, which involves updating a neural network's parameters through a process of backpropagation to minimize the loss function with respect to a new dataset. Fine-tuning a model with a number of parameters in the order of tens of billions, as is the case with modern language models, requires massive memory reserves merely to store optimizer states, gradients, and updated model parameters, effectively limiting this critical research to organizations with substantial cloud computing budgets.

3.3 Research Objectives and Contributions

- This research project aims to introduce a new framework called IndustryGPT that is both scalable and extremely resource-friendly with the specific purpose of building a conversational AI agent geared towards the IT industry. The main aim of this project is to fill the current gap between NLG and its technical applicability in the industry by using a pre-trained causal language model and only fine-tuning it on technical Q&A data. To ensure the absolute feasibility of this project with the hardware constraints of a standard machine that has a fixed constraint of only using a single NVIDIA T4 GPU with a fixed VRAM of only 16GB, this research uses a technique called Low Rank Adaptation (LORA). LoRA is a technique that reduces the number of trainable parameters in a pre-trained model by using the pre-trained weights and only introducing trainable rank decomposition matrices into the transformer architecture.
- The main contributions of this research paper have been clearly outlined as: Developing a data pipeline that is capable of extracting raw web data from the Stack Overflow website using advanced multiprocessing techniques. Implementing a PEFT strategy using a causal language model that is extremely resource-friendly. A qualitative evaluation of the conversational AI agent using LoRA that successfully demonstrates the unique effectiveness of the technique in such extremely constrained computational environments.

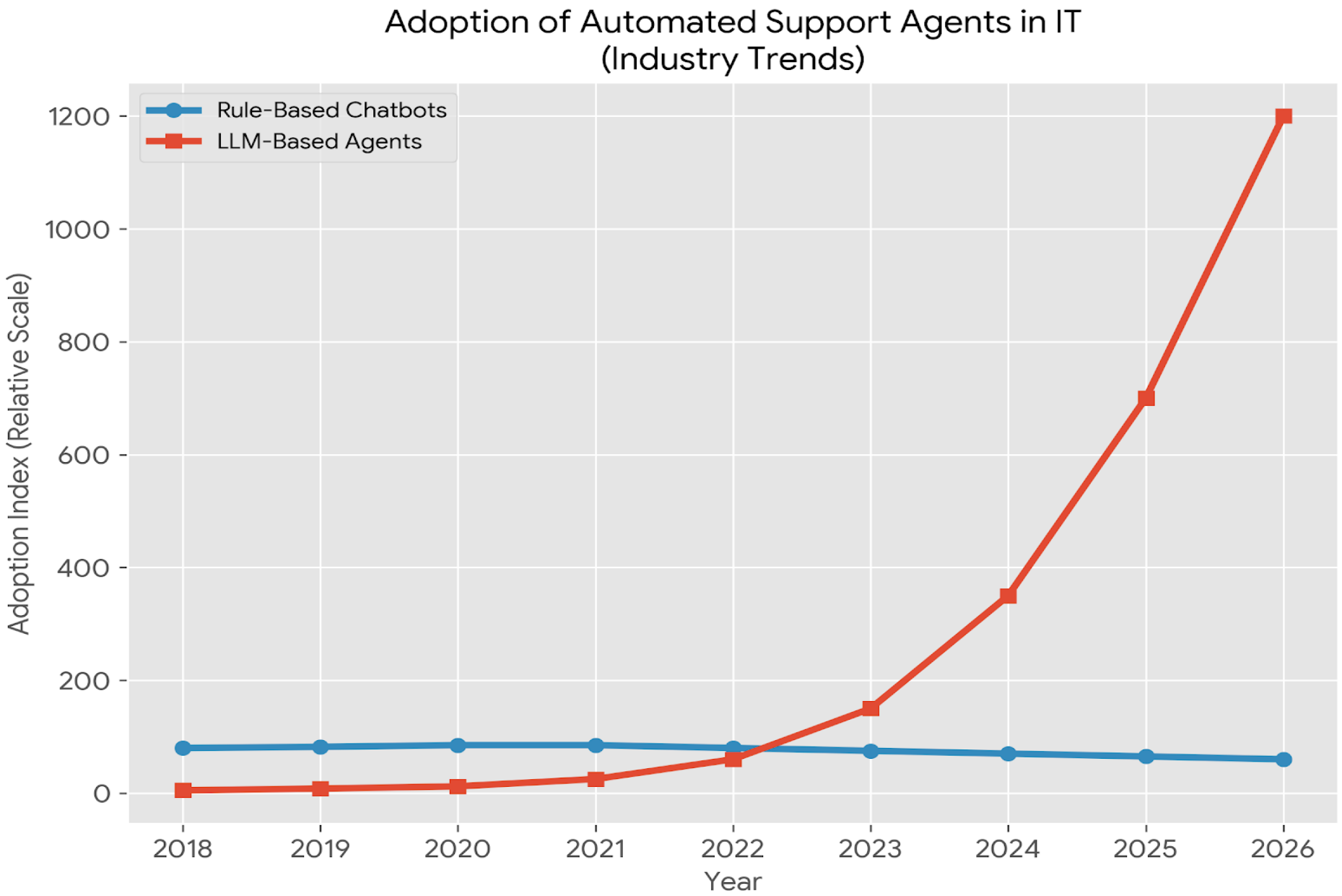
4. Industry Analysis

4.1 Industry Overview

- Currently, the Information Technology industry and software engineering operate in a complex environment where the pace of complexity is increasing exponentially. The architecture of the software and the methodology of the special AI assistant operate on the foundation of the critical convergence of machine learning research in the areas of natural language processing in software engineering, model optimization, and the development of automated support agents. The evolution of Information Technology support has fundamentally shifted from static knowledge bases that were often easily outdated to dynamic conversational support agents. The architecture of the frameworks supporting Information Technology service desks has traditionally been focused on rule-based chatbots with the requirement of explicit decision trees. These systems were inflexible and unable to support the variable complexity of modern software debugging.

4.2 Current Trends

- However, the adoption of Large Language Models in software engineering has accelerated exponentially. The foundational models of the LLM paradigm, such as OpenAI's Codex, GitHub Copilot, and Meta's Code LLAMA, have shown exceptional capabilities in the complex areas of software engineering. These models operate on the foundation of their training on enormous datasets of publicly available code bases and vast technical forums. The LLM paradigm has shown the capability of supporting the orchestration of complex multi-step operations and the provision of conversational debugging support.


Adoption of Automated Support Agents in IT
(Industry Trends)
Rule-Based Chatbots
LLM-Based Agents
Adoption Index (Relative Scale)
0
200
400
600
800
1000
1200
2018
2019
2020
2021
2022
2023
2024
2025
2026
Year

4.3 Market Dynamics and Implications for Research

- A specific market niche, however, persists in the realm of localized, highly specific enterprise implementations. Researchers have consistently identified a need to address the problem of using these enormous, generalized models to provide highly specific IT service management or developer support, which often results in unhelpful, generalized advice. It is well understood that "grounding" a model on highly specific, targeted data, such as questions on Stack Overflow, can dramatically improve the semantic accuracy of the model's output. This targeted approach ensures that the agent's "understanding" is in line with the actual syntax, terminology, and methodologies used by professional engineers.

4.4 Industry Requirements

- It is evident, therefore, that the current need in the IT industry is to create highly specific, scalable, secure, and conversational agents. IndustryGPT directly addresses this need, providing a highly specific, lightweight, internally fine-tuned conversational agent. A vital aspect of this is the agent's ability to utilize internally fine-tuned weights, avoiding the need to query a highly latency-intensive external API to provide fast, offline-capable responses to highly sensitive IT-specific questions. This provides superior, specialised accuracy while reducing data privacy concerns.

5. Review of Literature
5.1 Technical Domains with Large Language Models

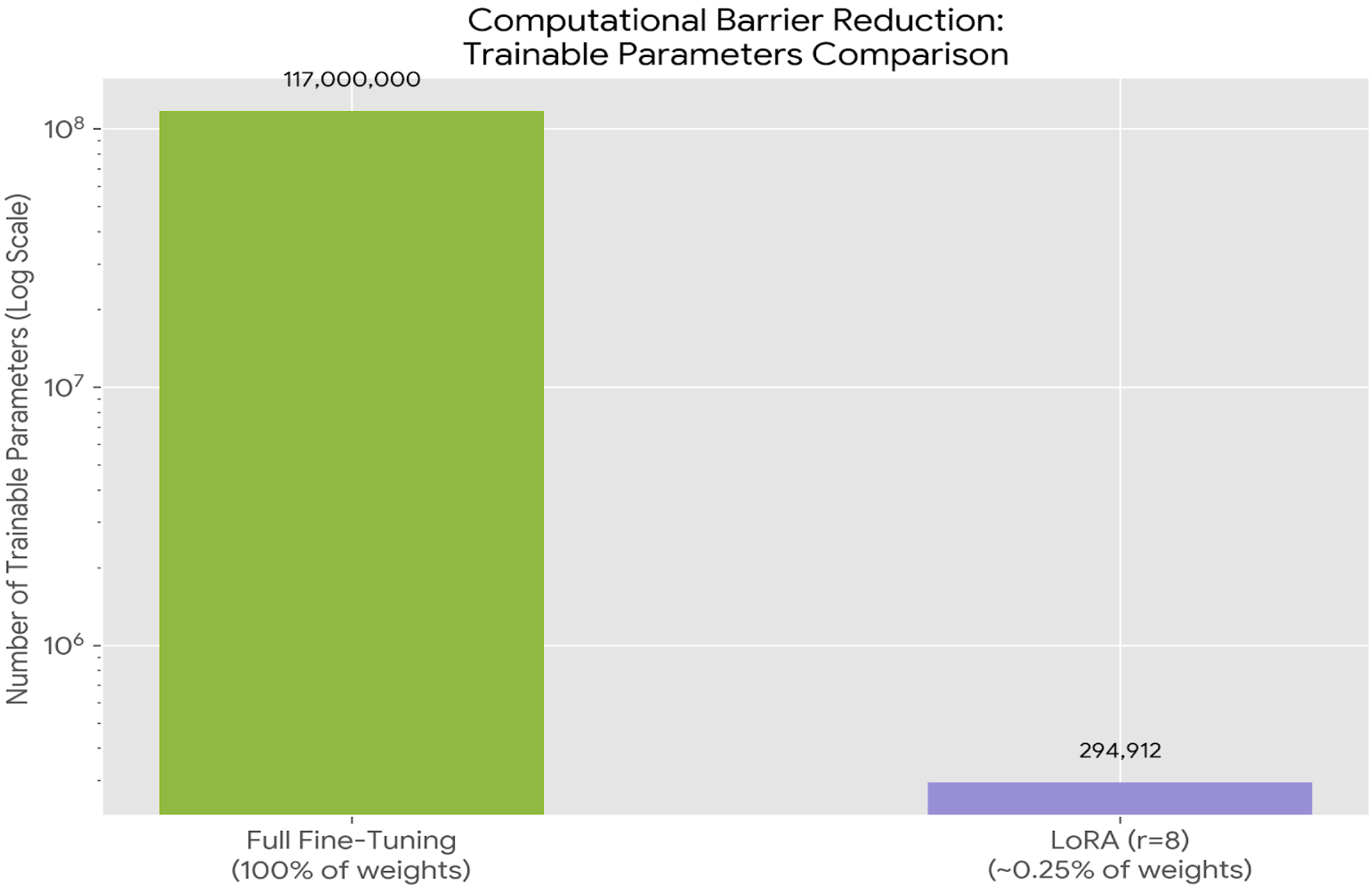
- Large Language Models' zero-shot reasoning capabilities are highlighted in the foundational literature, but using these models in specialised domains reveals important gaps. Although models trained on large codebases perform exceptionally well on generalised programming tasks, their application in particular operational environments frequently produces generic, shallow responses. A tried-and-true technique for aligning an LLM with exact engineering syntax and problem-solving techniques is grounding models using domain-specific data.

5.2 The Full Fine-Tuning Computational Barrier

- In the past, domain adaptation was accomplished by full model fine-tuning, which involves backpropagating all of a neural network's parameters. Complete fine-tuning is computationally prohibitive for contemporary LLMs with billions of parameters. According to the literature, this process necessitates enormous memory footprints in order to store optimiser states, gradients, and model weights. This effectively limits this cutting-edge research to large enterprises with substantial budgets for cloud computing.

5.3 LoRA and Parameter-Efficient Fine-Tuning (PEFT)

- The natural language processing (NLP) community has been actively looking for ways to make model training more accessible as model sizes have increased dramatically throughout the industry. The term "Parameter-Efficient Fine-Tuning" (PEFT) refers to a set of methods created especially to modify large models without changing the entire parameter space. Prefix-tuning, prompt tuning, and adapter layer injection are well-documented techniques.
- Low-Rank Adaptation (LoRA), which was formally introduced by Hu et al. (2021), has quickly emerged as the industry standard for effective model adaptation among these new approaches. The fundamental mathematical tenet of LoRA is that the "intrinsic rank" of the weight change needed for model adaptation is remarkably low. The following is how the theoretical framework functions: A mathematically smaller, trainable low-rank decomposition matrix is introduced into the architecture during the fine-tuning training loop, while the pre-trained weight matrix, represented as W_0, is completely frozen. The transformer's forward pass is then altered so that: $h = W_0 x + \Delta W x = W_0 x + B A x$
- The trainable low-rank matrices in this equation are denoted by B and A. The total number of trainable parameters is reduced by several orders of magnitude using this highly optimised method. By doing this, it successfully reduces the occurrence of catastrophic forgetting and permits thorough fine-tuning on common, consumer-grade hardware.


Computational Barrier Reduction:
Trainable Parameters Comparison
117,000,000
294,912
10^8
10^7
10^6
Number of Trainable Parameters (Log Scale)
Full Fine-Tuning
(100% of weights)
LoRA (r=8)
(~0.25% of weights)

5.4 Automated Developer Support Agents

- The field of PEFT approaches and automated support systems is one that is growing quickly. Rule-based chatbots were a major component of earlier IT support frameworks. By combining the PEFT framework with a highly targeted IT corpus, IndustryGPT offers a novel approach to this field and produces an offline-capable, internally optimised agent.

6. Techniques

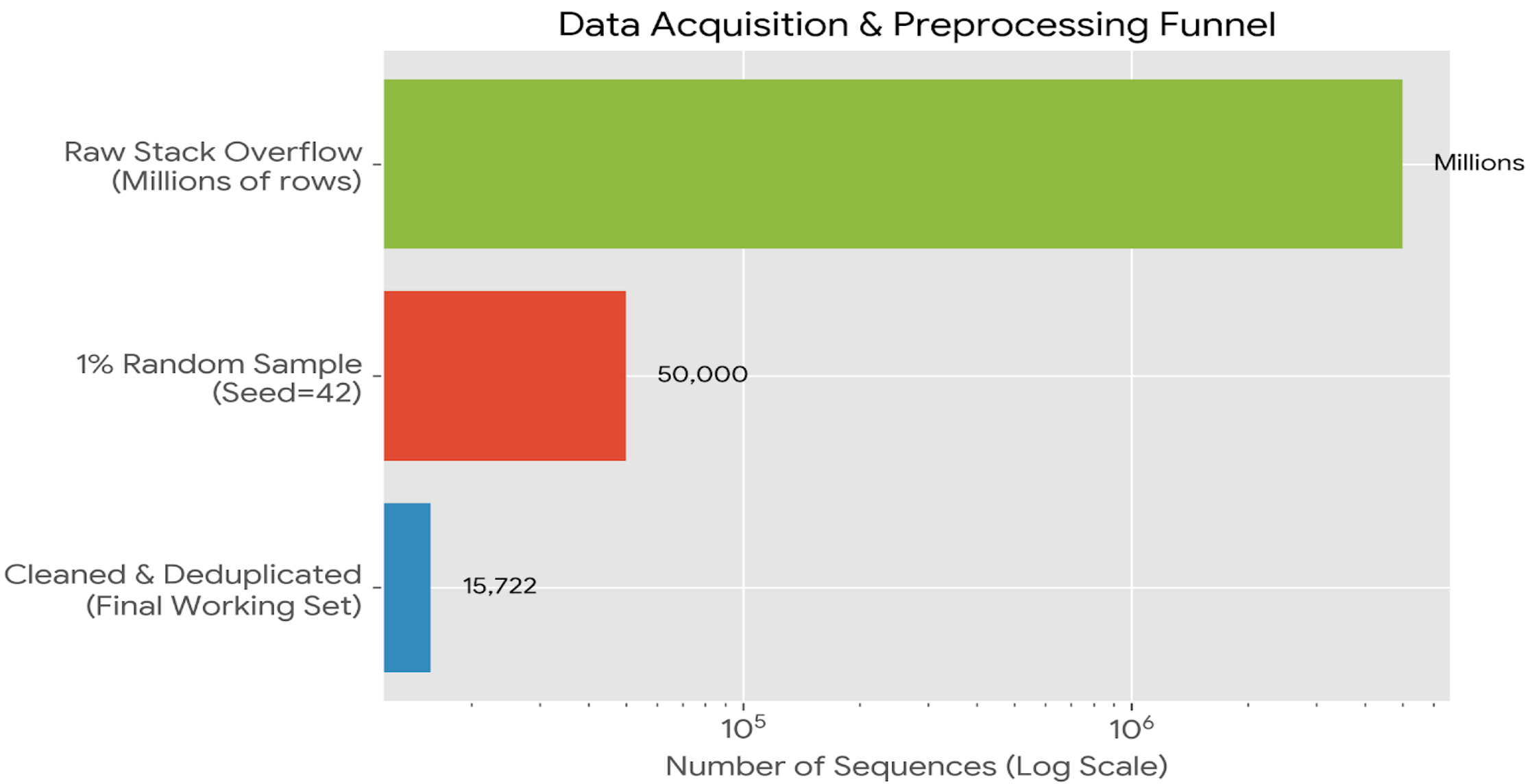
- IndustryGPT is developed using a highly structured, repeatable machine learning pipeline that includes tokenisation, prompt engineering, rigorous data preprocessing, model selection, and targeted Low-Rank Adaptation (LoRA) configuration.

6.1 Selection of Foundational Architecture

- The gpt2 architecture was chosen as the fundamental baseline for this study. OpenAI created GPT-2, a transformer-based causal language model made up of stacked decoder blocks. It is still a strong, well-documented architecture that is perfect for showcasing the effectiveness of PEFT pipelines without adding needless architectural overhead, even though it is smaller than more recent models. Importantly, it is naturally suited for multi-turn conversational Q&A tasks due to its auto-regressive nature, which is the mathematical process of predicting the next token in a sequence based on all previous tokens.

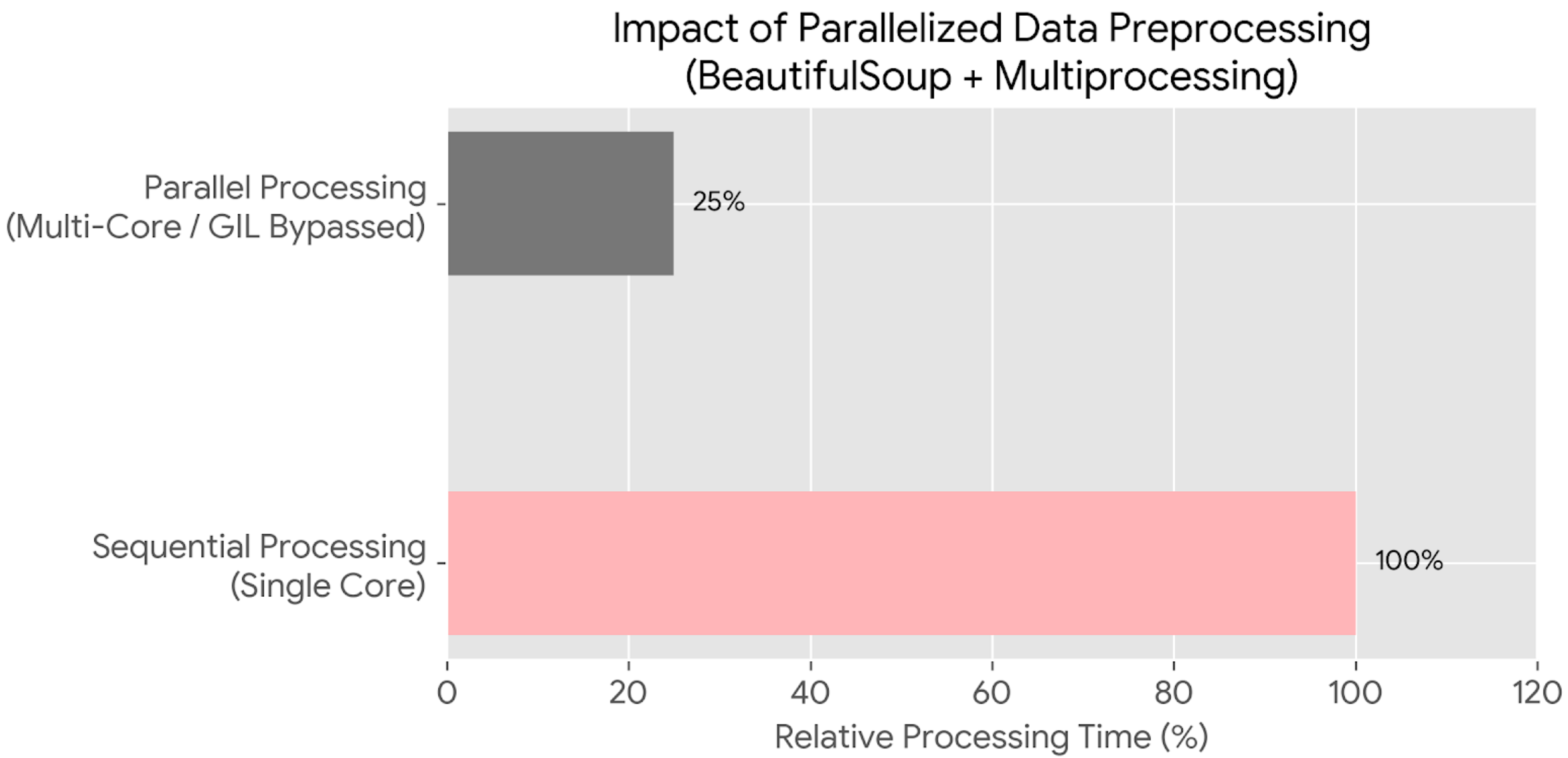
6.2 Memory-Constrained Sampling and Dataset Acquisition

- The pacovaldez/stackoverflow-questions dataset, safely stored on the Hugging Face Hub, is the source of the fundamental knowledge corpus for IndustryGPT. As the leading community-driven Q&A site for software developers, Stack Overflow provides an unmatched collection of real-world technical problems, debugging scenarios, and architectural conversations.
- However, with millions of text sequences, the entire Stack Overflow dataset is incredibly large. When attempting to load, preprocess, and train on the complete dataset in a typical cloud notebook environment, system RAM exhaustion often causes catastrophic "runtime disconnected" errors. A randomised 1% sample of the training set was taken in order to effectively address these memory limitations and guarantee a stable development lifecycle. To ensure deterministic reproducibility for subsequent researchers, a fixed seed (Seed = 42) was used throughout the sampling procedure. A very manageable working dataset of precisely 15,722 samples was produced as a result of this extraction. This particular sample size gave the model enough vocabulary and technical context to learn domain-specific patterns without going over the development environment's stringent 12GB system RAM limit.


Data Acquisition & Preprocessing Funnel
Raw Stack Overflow
(Millions of rows)
Millions
1% Random Sample
(Seed=42)
50,000
Cleaned & Deduplicated
(Final Working Set)
15,722
10^5
10^6
Number of Sequences (Log Scale)

6.3 Pipeline for Parallelised Preprocessing

- The HTML markup, code blocks, and formatting tags that are naturally present in raw data scraped from online forums cause significant noise during the tokenisation process. In order to prevent the model from learning HTML structural tags as natural language, text purification was an essential precondition. Using the Python BeautifulSoup library, we created a custom text-cleaning function that methodically parses and extracts raw text from intricate HTML structures.
- We developed a highly optimised parallel processing architecture because parsing more than 15,000 HTML strings sequentially has a substantial computational overhead. The dataset was divided into distinct sections that precisely matched the total number of available CPU cores using Python's built-in multiprocessing module. The Global Interpreter Lock (GIL) restrictions for I/O bound tasks were circumvented by mapping the cleaning function across these chunks concurrently, which significantly decreased preprocessing time and optimised hardware utilisation. The dataset was rigorously cleaned of any null values (NaNs) and duplicate entries in both the "title" (which represents the user question) and "body" (which represents the technical answer) features after text extraction.


Impact of Parallelized Data Preprocessing
(BeautifulSoup + Multiprocessing)
Parallel Processing
(Multi-Core / GIL Bypassed)
25%
Sequential Processing
(Single Core)
100%
0
20
40
60
80
100
120
Relative Processing Time (%)

6.4 Quick Engineering and Tokenisation Approach

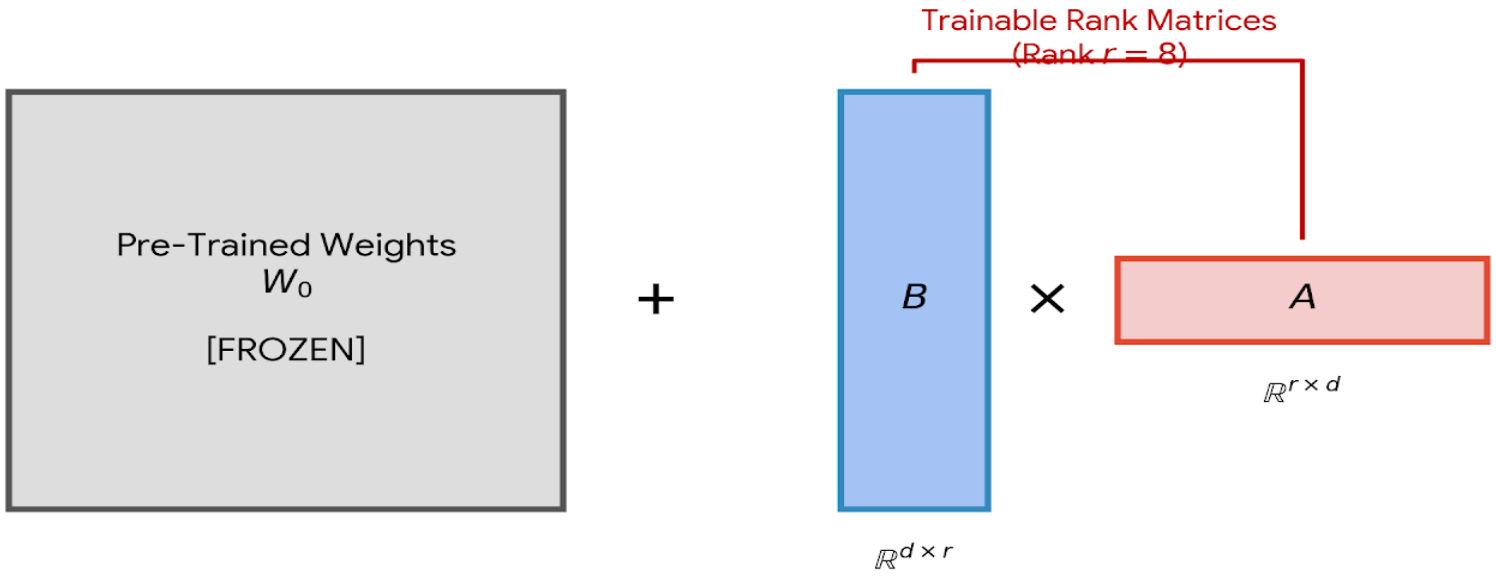
- The training dataset was mapped into a highly specific prompt template in order to change the GPT-2 model from a general text continuation generator to a structured conversational format. The exact format for each data point was as follows: ### Question: [Stack Overflow Title] ### Response: [Stack Overflow Body] This stringent formatting clearly indicates to the model's attention mechanism where the generated response should start and where the user query definitively ends. The AutoTokenizer class linked to the pre-trained GPT-2 model weights was then used to process the text sequences. All sequences were padded and truncated to a strict maximum length of 128 tokens in order to optimise GPU memory mapping and guarantee consistent tensor dimensions during batch processing. To facilitate the training loop, we designated the **End-Of-Sequence (EOS)** token as our padding marker. We also copied the input IDs directly into the labels tensor—a standard requirement for calculating loss in PyTorch when training language models.

6.5 Fine-Tuning Strategy (PEFT)

- The heart of our approach relies on **Parameter-Efficient Fine-Tuning (PEFT)** using the Hugging Face peft library. Instead of retraining the entire model, we "froze" most of the GPT-2 structure and only allowed specific sub-modules to adapt.
- We used a technique called **LoRA** (Low-Rank Adaptation) with a few strict settings:
- **Rank (r) of 8:** This determines the size of the new, smaller matrices we're adding to the model. We chose a rank of 8 because it hits the "sweet spot"—it keeps the number of trainable parameters incredibly low but still gives the model enough "brainpower" to learn specialized IT and tech vocabulary.
- **Alpha (α) of 32:** This is a scaling factor that controls how much weight the new LoRA updates carry during the model's thought process.
- **Target Modules:** We specifically pointed the updates at ["c_attn", "c_proj", "c_fc"]. By focusing on the attention mechanisms and the fully connected layers, we were able to upgrade the model’s reasoning and how it maps meanings, all while leaving its foundational "knowledge" of language untouched.
- **Dropout (0.1):** Think of this as a "sanity check" for the model. We applied it to the LoRA layers to keep the network on its toes. Since we were working with a small 1% data sample, dropout prevents the model from just memorizing the patterns (overfitting) and forces it to actually learn the underlying concepts.

Low-Rank Adaptation (LoRA) Architecture Forward Pass

$$h = W_0 x + BAx$$



7. Results and Qualitative Evaluation

- When you're building a conversational AI, you can't just look at a simple accuracy score. Instead, we have to look at the "human" side: Does the bot make sense? Is it staying on topic? Does it actually sound like it knows what it’s talking about in the IT world?

7.1 The "Budget" Lab Setup

- We ran the entire fine-tuning process in a cloud-hosted Jupyter Notebook using a single **NVIDIA T4 GPU**. Since this is a consumer-grade card with only 16GB of memory, we had to get creative to make the training work:
- **Small Batches, Big Results:** We could only fit 2 sequences on the card at a time. To prevent the model's learning from becoming erratic, we used **Gradient Accumulation**. By stacking up 4 small steps before updating the weights, we "tricked" the model into thinking it was working with a much more stable batch of 8.
- **Mixed-Precision (FP16):** We told the GPU to handle its math in 16-bit rather than the usual 32-bit. This effectively cut our memory usage in half and let the T4’s "Tensor Cores" sprint through the calculations.

7.2 Training Logic

The model went through the data 5 times (**5 epochs**). We used the **AdamW optimizer** with a standard "gentle" learning rate. We also added **L2 weight decay**, which acts like a penalty for over-complicating things. This forced the model to generalize broad IT concepts rather than just rote-memorizing our 15,000 training lines.

7.3 Making the Bot Converse

To make sure the bot didn't just spit out the same robotic answer every time, we used **Nucleus Sampling**. Here's how we tuned its "personality":

Max Tokens (150): Keeps answers concise so it doesn't start rambling.

Top-P & Top-K: These filters keep the bot focused on logical, high-probability words while ignoring the "weird" or irrelevant ones.

Temperature (0.7): This is the "creativity dial." We set it to 0.7 to get a perfect mix of factual structure and natural-sounding language.

7.4 Case Studies: Putting IndustryGPT to the Test

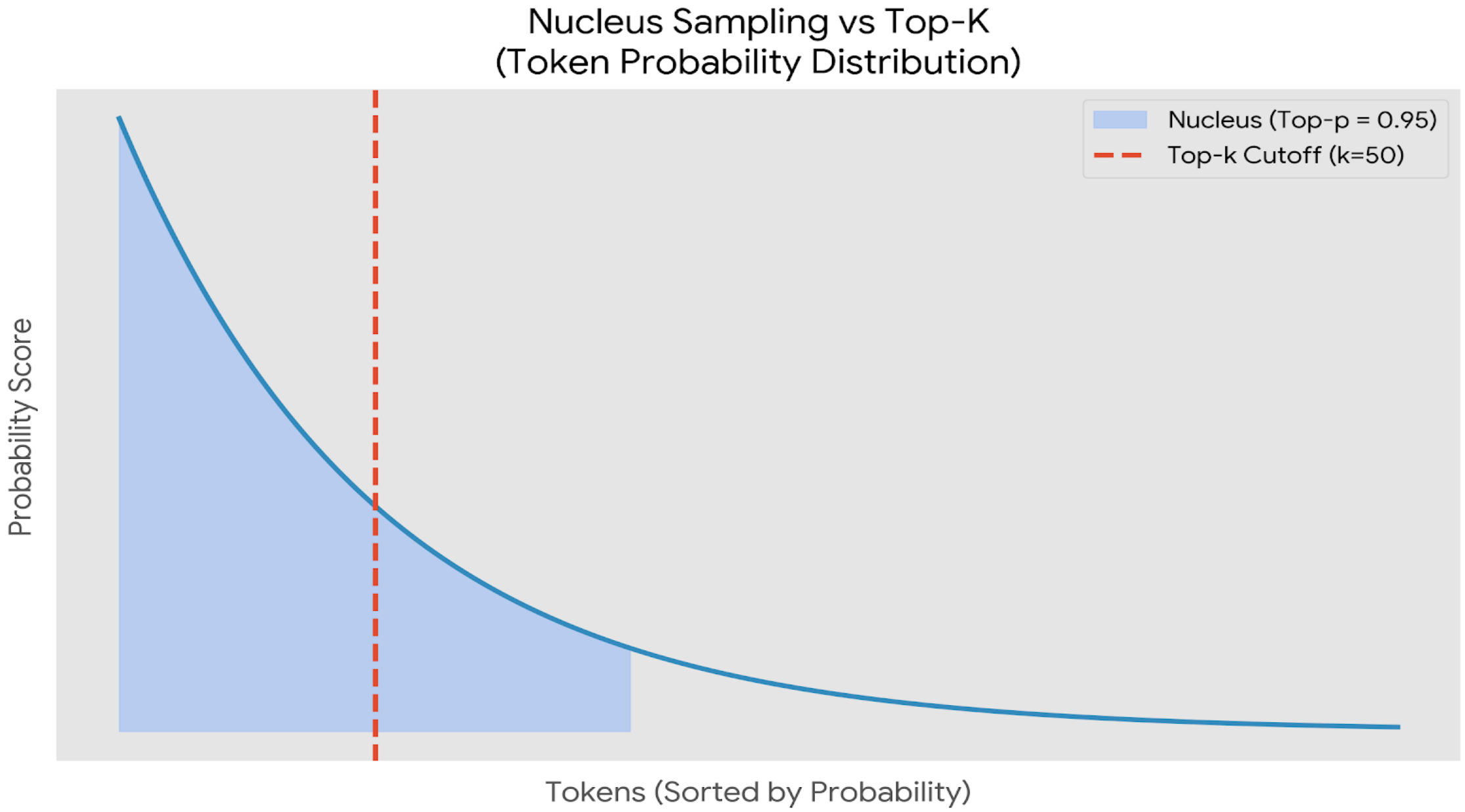
We asked the model questions it had never seen before to see if it truly "understood" IT:

SQL Optimization: Instead of saying "make it fast," the bot talked specifically about SQL Server structures and how large datasets are ingested.

DevOps/Containers: It correctly explained that containerization (like Docker) is about isolating apps from the OS to keep environments consistent.

AI Definitions: It clearly explained that Machine Learning is the "parent" category and Deep Learning is a specialized "subset."

APIs: It recognized the difference between old data methods and modern, stateless REST principles.


Nucleus Sampling vs Top-K
(Token Probability Distribution)
Nucleus (Top-p = 0.95)
Top-k Cutoff (k=50)
Probability Score
Tokens (Sorted by Probability)

8. Discussion and Limitations

- IndustryGPT shows that you don't need a supercomputer to build a specialized AI. By using **LoRA** and **Mixed-Precision**, we proved that independent researchers can build high-quality tools on a budget.
- Where we can improve:
- **The Data Gap:** Because we only used 1% of the data, the bot is a "Generalist Expert." It understands *what* SQL is, but it might struggle to write a 500-line complex script from scratch.
- **The "Brain" Size:** GPT-2 is a classic, but it has a small "context window." It can't "remember" a 20-page document you paste into the chat.
- **Concept vs. Code:** The bot is currently a great teacher, but it needs more scale to become a perfect, line-by-line coding assistant.
- What’s Next?
- **QLoRA:** Using even bigger models (like LLaMA-2) but "compressing" them so they still run on cheap hardware.
- **RAG (Retrieval-Augmented Generation):** Instead of the bot relying only on its memory, we can give it a "digital library" to look up the latest tech manuals in real-time. This would virtually eliminate the chance of it making things up (hallucinating).

9. Conclusion

- This project is a win for accessible AI. We proved that with the right engineering—like parallel data cleaning and smart memory management—you can turn a generic model into a specialized **IndustryGPT**. It shows that secure, domain-specific AI isn't just for big tech companies; it's something a dedicated researcher can build in their own home office.

10. References

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Appendix A
Parallelized Data Preprocessing Script

- Real-world web data is notoriously messy. When scraping forums like Stack Overflow, you inherit a lot of HTML tags and formatting noise that can confuse a language model.
- This script shows how we built our data extraction and cleaning pipeline. To keep things moving fast, we combined BeautifulSoup (to strip away the HTML junk) with Python's multiprocessing library (to divide the workload across all available CPU cores concurrently).
- Python
- import multiprocessing
- import pandas as pd
- import numpy as np
- from bs4 import BeautifulSoup
- from datasets import load_dataset

```
# 1. Load the dataset and extract a 1% reproducible sample
print("Loading dataset...")
full_dataset = load_dataset("pacovaldez/stackoverflow-questions", split="train")
sampled_dataset = full_dataset.shuffle(seed=42).select(range(int(len(full_dataset) * 0.01)))
# Convert to Pandas DataFrame for easier multiprocessing mapping
df = sampled_dataset.to_pandas()
# 2. Define the HTML cleaning function
def clean_html_content(raw_html):
"""Parses and extracts raw text from HTML structures."""
if pd.isna(raw_html):
return ""
soup = BeautifulSoup(raw_html, "html.parser")
return soup.get_text(separator=" ", strip=True)
# 3. Parallelized Processing Architecture
def parallelize_dataframe(df, func, num_cores):
"""Splits the dataframe into chunks and processes them concurrently."""
df_split = np.array_split(df, num_cores)
pool = multiprocessing.Pool(num_cores)
df = pd.concat(pool.map(func, df_split))
pool.close()
pool.join()
return df
```

```
def process_chunk(chunk):
"""Applies the cleaning function to the 'body' feature of a chunk."""
chunk['cleaned_body'] = chunk['body'].apply(clean_html_content)
return chunk
if __name__ == '__main__':
# Determine available CPU cores
num_cores = multiprocessing.cpu_count()
print(f"Starting parallel cleaning on {num_cores} cores...")
# Executing parallel cleaning
cleaned_df = parallelize_dataframe(df, process_chunk, num_cores)
# Purge null values and drop duplicates across title and body

cleaned_df.dropna(subset=['title', 'cleaned_body'], inplace=True)
cleaned_df.drop_duplicates(subset=['title', 'cleaned_body'], inplace=True)
print(f"Preprocessing complete. Final sample size: {len(cleaned_df)}")
```

Appendix B: LoRA Configuration and Training Loop

- This is where the heavy lifting happens—without melting our hardware. Instead of running a full fine-tuning loop that updates every single weight in the GPT-2 model (which requires massive cloud infrastructure), we used Parameter-Efficient Fine-Tuning (PEFT).
- The code below sets up our Low-Rank Adaptation (LoRA) strategy. It freezes the foundational model and only injects a small set of trainable parameters into the attention and fully connected layers.
- Python
- import torch
- from transformers import AutoModelForCausalLM, AutoTokenizer, TrainingArguments, Trainer
- from peft import LoraConfig, get_peft_model
- # 1. Initialize Base Model and Tokenizer
- model_name = "gpt2"
- tokenizer = AutoTokenizer.from_pretrained(model_name)
- tokenizer.pad_token = tokenizer.eos_token # Designate EOS as padding token
- model = AutoModelForCausalLM.from_pretrained(model_name)

```
# 2. Configure Low-Rank Adaptation (LoRA)
lora_config = LoraConfig(
r=8, # Dimension of low-rank matrices
lora_alpha=32, # Scaling factor
target_modules=["c_attn", "c_proj", "c_fc"], # Target attention and fully connected layers
lora_dropout=0.1, # Regularization
bias="none",
task_type="CAUSAL_LM"
)
# Inject LoRA matrices into the frozen base model
peft_model = get_peft_model(model, lora_config)
peft_model.print_trainable_parameters()
# 3. Define Resource-Constrained Training Arguments
training_args = TrainingArguments(
output_dir="./IndustryGPT-results",
num_train_epochs=5,
per_device_train_batch_size=2, # Strict VRAM limitation
gradient_accumulation_steps=4, # Simulated batch size of 8
learning_rate=5e-5,
weight_decay=0.01, # L2 penalty
fp16=True, # Mixed-Precision Training
```

```
# 4. Initialize Trainer
trainer = Trainer(
model=peft_model,
args=training_args,
train_dataset=tokenized_dataset, # Assuming dataset was previously tokenized
data_collator=lambda data: {
'input_ids': torch.stack([f[0] for f in data]),
'attention_mask': torch.stack([f[1] for f in data]),
'labels': torch.stack([f[0] for f in data]) # input_ids copied to labels
}
)
# Execute Fine-Tuning
trainer.train()
```

Appendix C: Inference Generation Script

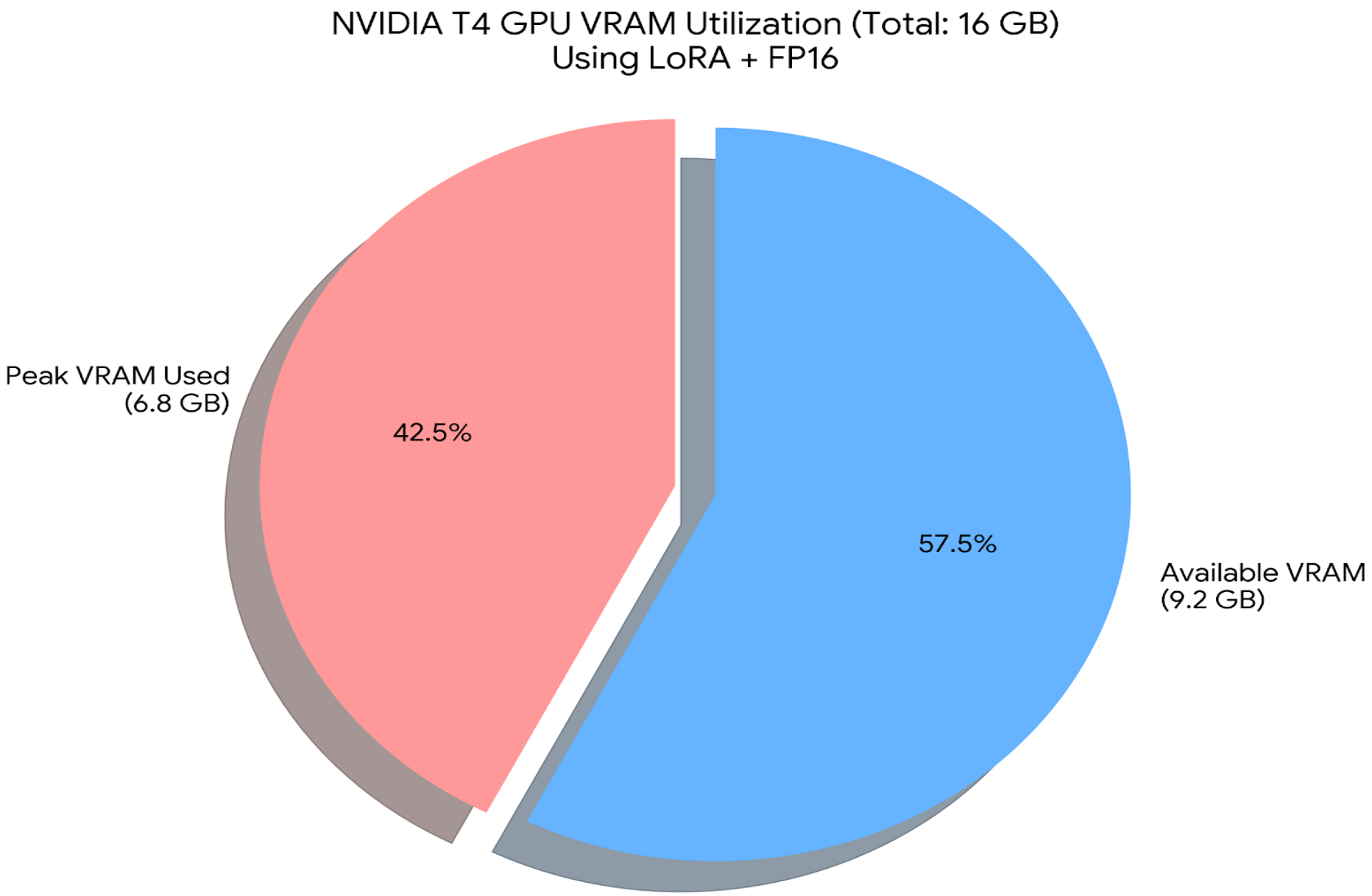
- Once the model is trained, it needs to be able to actually converse. This script sets up the pipeline we used to test the bot's qualitative performance.
- To keep the bot from sounding robotic or getting stuck in repetitive loops, we tuned the "nucleus sampling" parameters (like temperature and top-p). This ensures the responses hit the right balance between strict factual accuracy and creative, natural-sounding generation.
- Python
- from transformers import pipeline
- # Load the fine-tuned PEFT model
- generator = pipeline(
- "text-generation",
- model=peft_model,
- tokenizer=tokenizer,
- device=0 if torch.cuda.is_available() else -1
-)

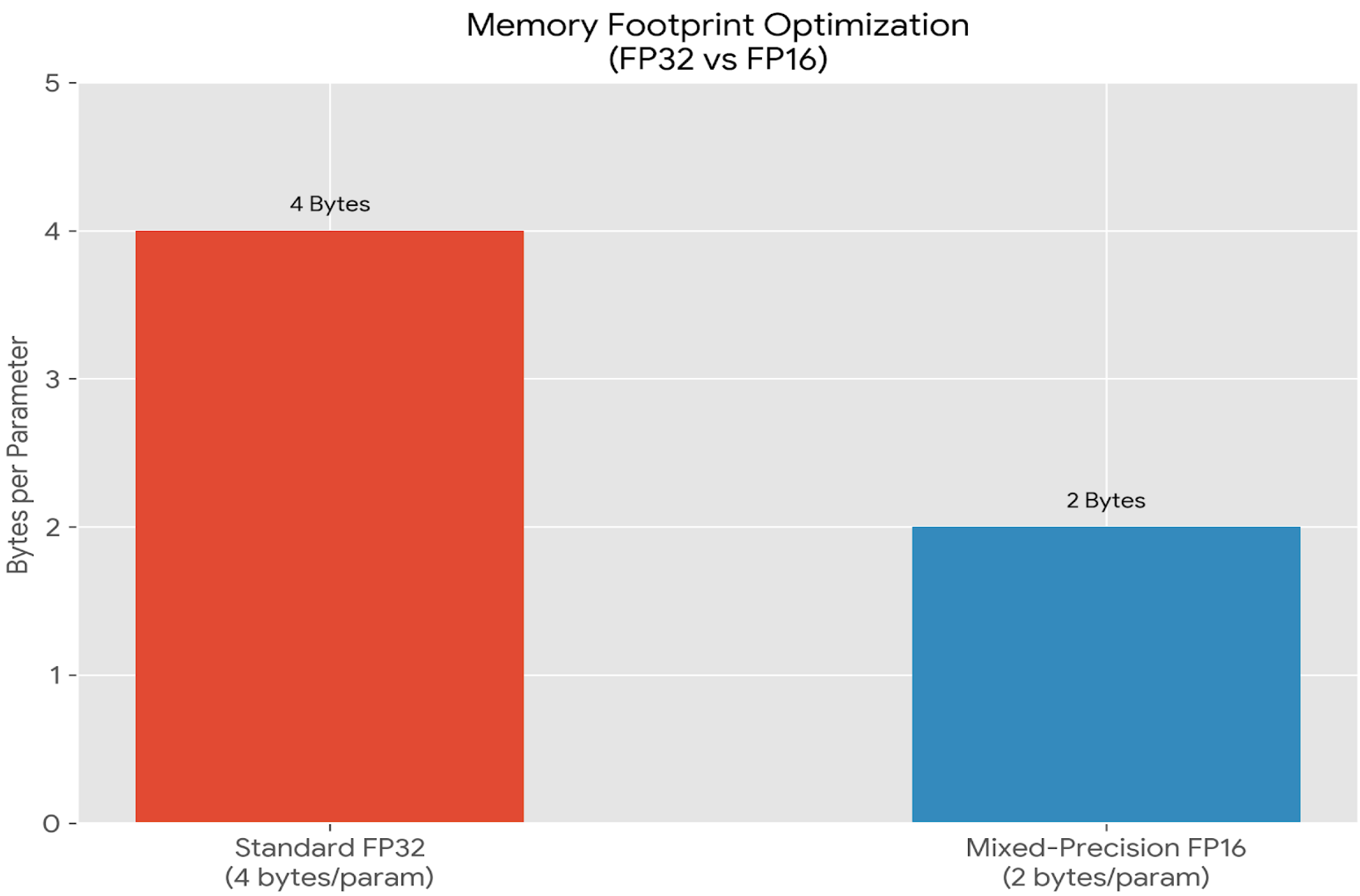
```
def ask_industrygpt(question):
"""Formats the prompt and generates a response using configured
sampling."""
prompt = f"### Question: {question}\n### Answer:"
response = generator(
prompt,
max_new_tokens=150,
top_p=0.95, # Nucleus sampling
top_k=50, # Restrict to 50 most likely tokens
temperature=0.7, # Balance determinism with creative generation
pad_token_id=tokenizer.eos_token_id,
return_full_text=False
)

return response[0]['generated_text'].strip()
# Example Usage
print(ask_industrygpt("How to optimize a SQL query for large datasets?"))
```

Appendix D: Detailed Hardware and Memory Calculations

- Running out of VRAM is the bane of any ML engineer's existence. To prove that you can train an effective, specialized LLM without enterprise hardware, we tracked our memory constraints very carefully. Here is the math showing exactly how we stayed safely below the 16GB limit of a standard NVIDIA T4 GPU:
- **1. Effective Batch Size Simulation**
- If we tried to load a large batch of data directly onto the GPU, it would crash. So, we limited the physical batch size to 2 sequences at a time, but used gradient accumulation to "trick" the model into learning from a larger, more stable batch size of 8 before taking a step.
- $$Effective\ Batch\ Size = Physical\ Batch\ Size \times Gradient\ Accumulation\ Steps$$
- $$Effective\ Batch\ Size = 2 \times 4 = 8$$
- **2. Mixed-Precision VRAM Optimization**
- Standard models handle their math using 32-bit floating points (FP32), which takes up about 4 bytes per parameter. By switching our training loop to use 16-bit floating points (FP16), we only needed 2 bytes per parameter.
- This effectively cut our memory footprint in half. Combined with LoRA (which already froze roughly 99% of the base model's 117 million parameters), we kept our peak VRAM usage around **6.8 GB**, giving us plenty of breathing room on our 16GB card.


NVIDIA T4 GPU VRAM Utilization (Total: 16 GB)
Using LoRA + FP16
Peak VRAM Used
(6.8 GB)
42.5%
57.5%
Available VRAM
(9.2 GB)


Memory Footprint Optimization
(FP32 vs FP16)
Bytes per Parameter
5
4
3
2
1
0
4 Bytes
2 Bytes
Standard FP32
(4 bytes/param)
Mixed-Precision FP16
(2 bytes/param)